

Carboprost Solution For Injection USP 250 mcg/mL

As this preparation contains benzyl alcohol, its use shall be avoided in children under 2 years of age. Not to be used in neonates.

DESCRIPTION

Clear, colourless solution free from visible particulate matter.

Each vial contains Carboprost Tromethamine equivalent to 250 mcg Carboprost.

CLINICAL PHARMACOLOGY

Carboprost tromethamine administered intramuscularly stimulates in the gravid uterus myometrial contractions similar to labor contractions at the end of a full term pregnancy.

Whether or not these contractions result from a direct effect of carboprost on the myometrium has not been determined. Nonetheless, they evacuate the products of conception from the uterus in most cases.

Postpartum, the resultant myometrial contractions provide hemostasis at the site of placentation.

Carboprost tromethamine also stimulates the smooth muscle of the human gastrointestinal tract. This activity may produce the vomiting or diarrhea or both that is common when carboprost tromethamine is used to terminate pregnancy and for use postpartum. In laboratory animals and also in humans carboprost tromethamine can elevate body temperature. With the clinical doses of carboprost tromethamine used for the termination of pregnancy, and for use postpartum, some patients do experience transient temperature increases.

In laboratory animals and in humans large doses of carboprost tromethamine can raise blood pressure, probably by contracting the vascular smooth muscle. With the doses of carboprost tromethamine used for terminating pregnancy, this effect has not been clinically significant. In laboratory animals and also in humans carboprost tromethamine can elevate body temperature. With the clinical doses of carboprost tromethamine used for the termination of pregnancy, some patients do experience temperature increases. In some patients, carboprost tromethamine may cause transient bronchoconstriction.

Drug plasma concentrations were determined by radioimmunoassay in peripheral blood samples collected by different investigators from 10 patients undergoing abortion. The patients had been injected intramuscularly with 250 micrograms of carboprost at two hour intervals. Blood levels of drug peaked at an average of 2060 picograms/mL one-half hour after the first injection then declined to an average concentration of 770 picograms/mL two hours after the first injection just before the second injection. The average plasma concentration one-half hour after the second injection was slightly higher (2663 picograms/mL) than that after the first injection and decreased again to an average of 1047 picograms/mL by two hours after the second injection. Plasma samples were collected from 5 of these 10 patients following additional injections of the prostaglandin.

The average peak concentrations of drug were slightly higher following each successive injection of the prostaglandin, but always decreased to levels less than the preceding peak values by two hours after each injection.

Five women who had delivery spontaneously at term were treated immediately postpartum with a single injection of 250 micrograms of carboprost tromethamine. Peripheral blood samples were collected at several times during the four hours following treatment and carboprost tromethamine levels were determined by radioimmunoassay. The highest concentration of carboprost tromethamine was observed at 15 minutes in two patients (3009 and 2916 picograms/mL), at 30 minutes in two patients (3097 and 2792 picograms/mL), and at 60 minutes in one patient (2718 picograms/mL).

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INDICATIONS AND USAGE

Deviprost is indicated for aborting pregnancy between the 13th and 20th weeks of gestation as calculated from the first day of the last normal menstrual period and in the following conditions related to second trimester abortion:

1. Failure of expulsion of the fetus during the course of treatment by another method;
2. Premature rupture of membranes in intrauterine methods with loss of drug and insufficient or absent uterine activity;
3. Requirement of a repeat intrauterine instillation of drug for expulsion of the fetus;
4. Inadvertent or spontaneous rupture of membranes in the presence of a previable fetus and absence of adequate activity for expulsion.

DEVIPROST is indicated for the treatment of postpartum hemorrhage due to uterine atony which has not responded to conventional methods of management. Prior treatment should include the use of intravenously administered oxytocin, manipulative techniques such as uterine massage and, unless contraindicated, intramuscular ergot preparations.

Studies have shown that in such cases, the use of DEVIPROST has resulted in satisfactory control of hemorrhage, although it is unclear whether or not ongoing or delayed effects of previously administered ecbolic agents have contributed to the outcome. In a high proportion of cases, DEVIPROST used in this manner has resulted in the cessation of life threatening bleeding and the avoidance of emergency surgical intervention.

CONTRAINDICATIONS

1. Deviprost should not be used where the patient is sensitive to carboprost tromethamine or to any of the excipients
2. Acute pelvic inflammatory disease.
3. Patients with known active cardiac, pulmonary, renal, or hepatic disease.
4. Deviprost is contra-indicated in pregnancy.

WARNINGS

Deviprost (carboprost tromethamine), like other potent oxytocic agents, should be used only with strict adherence to recommended dosages. Deviprost should be used by medically trained personnel in a hospital which can provide immediate intensive care and acute surgical facilities.

Deviprost should be used by medically trained personnel and is available only to hospitals and clinics with specialised obstetric units where 24 hour resident medical cover is provided. Deviprost, as with other potent oxytocic agents, should be used only with strict adherence to recommended dosages.

This preparation should not be used for induction of labour.

Deviprost must not be given intravenously.

Special caution is necessary in patients with history of asthma, hypo- or hypertension, cardiovascular, renal, or hepatic disease, glaucoma or raised intra-ocular pressure, anaemia, jaundice, diabetes, or epilepsy.

Benefit/risk ratio should be assessed in patients with cardiovascular disease (risk of decreased blood pressure up to cardiovascular collapse, bradycardia), and in patients with a history of asthma (risk of bronchoconstriction) and pulmonary disease (possibility of decreased pulmonary blood flow and increased arterial pulmonary pressure).

Very rare cases of cardiovascular collapse have been reported following the use of prostaglandins. This

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should always be considered when using Deviprost.

Decreases in maternal arterial oxygen content have been observed in patients treated with carboprost tromethamine. A causal relationship to carboprost tromethamine has not been established, however, it is recommended that patients with pre-existing cardio-pulmonary problems receiving Deviprost are monitored during treatment and given additional oxygen if necessary.

As with any oxytocic agent, Deviprost should be used with caution in patients with previously compromised (scarred) uteri.

Prior treatment with, or concomitant administration of anti-emetics and antidiarrhoeal drugs significantly reduces the very high incidence of the gastrointestinal side effects common to all prostaglandins. Their use should be considered an integral part of the management of patients.

Transient pyrexia that may be due to hypothalamic thermoregulation has been observed after intramuscular Deviprost. Temperature elevations exceeding 1.1 °C were observed in approximately one-eighth of patients who received the recommended dosage regimen but if not complicated by endometritis, the temperature elevation will usually return to normal within several hours of the last injection.

Animal studies lasting several weeks at high doses have shown that prostaglandins of the E and F series can induce proliferation of bone. Such effects have also been noted in newborn infants who have received prostaglandin E1 during prolonged treatment. There is no evidence that short-term administration of Deviprost can cause similar bone effects.

Benzyl alcohol: This medicine contains 9.45 mg benzyl alcohol in each ampoule which is equivalent to 9.45 mg/ml. Benzyl alcohol may cause allergic reactions.

The preservative benzyl alcohol has been associated with serious adverse events, including the “gasping syndrome”, and death in paediatric patients. The minimum amount of benzyl alcohol at which toxicity may occur is not known. The risk of benzyl alcohol toxicity depends on the quantity administered and the liver and kidneys' capacity to detoxify the chemical. Premature and low-birth weight infants may be more likely to develop toxicity. High volumes should be used with caution and only if necessary, especially in subjects with liver or kidney impairment because of the risk of accumulation and toxicity (metabolic acidosis).

This medicine contains less than 1 mmol sodium (23 mg) per ml of solution, that is to say essentially 'sodium-free'

Drug Interactions

DEVIPROST may augment the activity of other oxytocic agents. Concomitant use with other oxytocic agents is not recommended.

Fertility

There are no data on the effects of carboprost on fertility

Pregnancy

Studies in animals have shown reproductive toxicity and any dose which produces increased uterine tone could put the embryo or foetus at risk. Benzyl alcohol can cross the placenta

Breast-feeding

There are no data on the excretion into breast milk for carboprost tromethamine

Pediatric Use

Safety and effectiveness in pediatric patients have not been established.

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Effects on ability to drive and use machines

No data on the effects on the ability to drive and use machines is available.

There have been reports of undesirable effects such as syncope, dizziness and somnolence which could impair the ability to drive or use machines.

Therefore patients should refrain from driving until they know that Deviprost does not affect their ability to drive or use machines.

ADVERSE REACTIONS

The table below lists the adverse effects identified through clinical trials and post-marketing surveillance by System Organ Class (SOC) and frequency. Within each frequency grouping, adverse events are presented in order of decreasing seriousness. Frequencies are defined as: very common, common, uncommon, rare, very rare, or not known.

The adverse effects of Deviprost are generally transient and reversible on discontinuation of therapy. The most frequent adverse reactions observed are related to its contractile effect on smooth muscles.

In the patients, some experienced vomiting and diarrhoea, some had nausea, some had a temperature increase greater than 1.1° C, and some experienced flushing.

MedDRA System Organ Class	Frequency	Undesirable Effects
Infections and Infestations	Uncommon	Septic shock, Urinary tract infection
	Common	Endometritis*
Immune system disorders	Not Known	Hypersensitivity reactions [†] (e.g. Anaphylactic reaction, Anaphylactic shock, Anaphylactoid reaction, Angioedema)
Endocrine disorders	Not Known	Thyrototoxic crisis [†]
Psychiatric disorders	Uncommon	Sleep disorder
	Not Known	Anxiety [†] , Nervousness [†]
Nervous system disorders	Common	Headache*
	Uncommon	Syncope vasovagal, Dizziness*, Dystonia, Paraesthesia, Somnolence, Dysgeusia, Lethargy
	Not Known	Syncope [†]
Eye disorders	Uncommon	Vision blurred, Eye pain
Ear and labyrinth disorders	Uncommon	Vertigo, Tinnitus
Cardiac disorders	Uncommon	Tachycardia
	Not Known	Palpitations [†]
Vascular disorders	Common	Flushing, Hot flush, Chills
	Uncommon	Hypertension
Respiratory, thoracic and mediastinal disorders	Common	Cough
	Uncommon	Asthma, Respiratory distress, Dyspnoea, Hyperventilation*, Wheezing, Hiccups
	Not Known	Bronchospasm, Pharyngeal oedema, Choking sensation [†] , Epistaxis [†] , Dry throat [†] , Upper respiratory tract infection
Gastrointestinal disorders	Very common	Diarrhoea*, Nausea*, Vomiting*
	Uncommon	Haematemesis, Abdominal pain upper, Dry mouth
	Not Known	Retching [†]
Skin and subcutaneous tissue disorders	Uncommon	Hyperhidrosis

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	Not Known	Rash [†]
Musculoskeletal and connective tissue disorders	Uncommon	Torticollis, Back pain, Myalgia,
	Not Known	Muscle spasms, Blepharospasm [†]
Reproductive system and breast disorders	Common	Uterine haemorrhage, Retained placenta or membranes
	Uncommon	Uterine rupture, Uterine cervical laceration, Pelvic pain*, Breast tenderness
	Not Known	Uterine disorder
General disorders and administration site conditions	Uncommon	Chest discomfort, Injection site pain
	Not Known	Chest pain [†] , Asthenia [†] , Excessive thirst [†]
Investigations	Very common	Body temperature increased

* Events reported for both intramuscular and intra-amniotic routes of administration are marked with an asterisk. All other events were reported only for the intramuscular route.

[†] Identified from post-marketing experience

DOSAGE AND ADMINISTRATION**1.Abortion and Indications 1–4**

An initial dose of 1 mL of Deviprost (containing the equivalent of 250 micrograms of carboprost) is to be administered deep in the muscle with a tuberculin syringe. Subsequent doses of 250 micrograms should be administered at 1½ to 3½ hour intervals depending on uterine response.

An optional test dose of 100 micrograms (0.4 mL) may be administered initially. The dose may be increased to 500 micrograms (2 mL) if uterine contractility is judged to be inadequate after several doses of 250 micrograms (1 mL). The total dose administered of carboprost tromethamine should not exceed 12 milligrams and continuous administration of the drug for more than two days is not recommended.

2.For Refractory Postpartum Uterine Bleeding

An initial dose of 250 micrograms of Deviprost (1 mL of DEVIPROST) is to be given deep, intramuscularly. In clinical trials it was found that the majority of successful cases (73%) responded to single injections. In some selected cases, however, multiple dosing at intervals of 15 to 90 minutes was carried out with successful outcome. The need for additional injections and the interval at which these should be given can be determined only by the attending physicians as dictated by the course of clinical events. The total dose of DEVIPROST should not exceed 2 milligrams (8 doses).

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

Overdose

Treatment of overdosage must be symptomatic and supportive as clinical studies with prostaglandin antagonists have not progressed to the point where recommendations may be made.

If evidence of excessive side-effects appears, the frequency of administration should be decreased or administration discontinued.

Shelf Life

24 Months

Storage conditions

Store in a refrigerator between 2° to 8°C. For Single Use Only. Discard the unused portion immediately after open

DEVIPROST



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Nature and contents of container

2 mL USP Type-I tubular flint glass vials and stoppered with 13 mm grey colored Teflon coated rubber stoppers and sealed with 13 mm blue color flip off seals containing 1mL solution.

10 vial per carton

Manufactured by:

Dr. Reddy's Laboratories Limited,

FTO-IX, Plot No's.Q1 to Q5,

Phase III, VSEZ,

Duvvada, Visakhapatnam District

530046, Andhra Pradesh, India

PRODUCT REGISTRATION HOLDER

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